

HAT-P-12b

R-1243

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_200413 · created 2026-07-31T11:11:28+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458952.8221 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.113 +/- 0.027
Transit depth (Rp/Rs)^2	1.28 +/- 0.62 [%]
Orbital Inclination (inc)	88.07 +/- 1.15
Airmass coefficient 1 (a1)	0.7352 +/- 0.009
Airmass coefficient 2 (a2)	0.283 +/- 0.08
Scatter in the residuals of the lightcurve fit is	1.23 %
Best Comparison Star	#2 - [599, 131]
Optimal Method	PSF photometry
Transit Duration (day)	0.0878 +/- 0.0097

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.113 ± 0.027 vs 0.1406 (deviation 1.02σ)
Duration fit vs published	0.0878 d vs 0.0975 d (deviation 1.0σ)
Mid-time O-C	1.7 min
Depth SNR	4.2
Residual scatter	1.23 %

Light curve

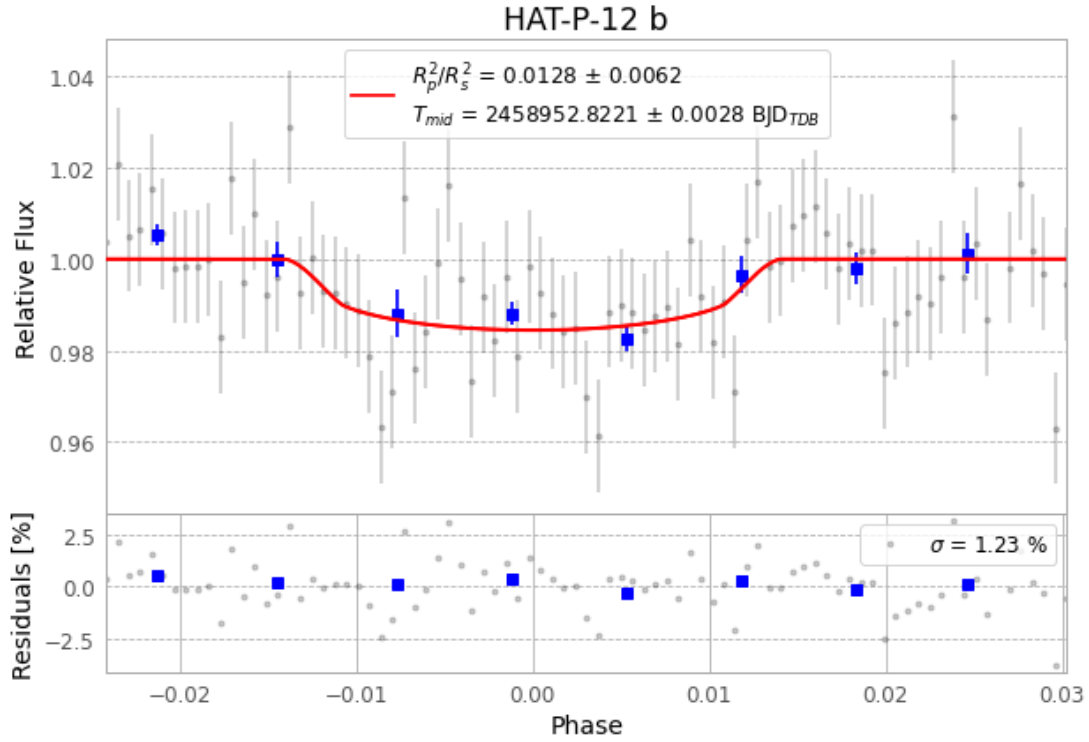


Figure 1 — FinalLightCurve_HAT-P-12 b_2020-04-12.png (SHA-256 082458d99ea0c532...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [529, 240], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d68d014ee9fdc294...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

85 FITS frames (56 MB), HATP-12200413054512.FITS → HATP-12200413095712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-200413.log (185c87636f77...), FinalLightCurve_HAT-P-12 b_2020-04-12.png (082458d99ea0...), FinalLightCurve_HAT-P-12 b_2020-04-12.csv (d6693c3afbb9...)

Compute resources

Wall time 170.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-31 11:11Z.